SUPPLEMENTARY MATERIAL: DIACHRONIC MODELING OF TONAL COHERENCE ON THE TONNETZ ACROSS CLASSICAL AND POPULAR REPERTOIRES

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A. CORPUS

A.1 Overview

Table 1 summarizes the two corpora used in this study.

Table 1: Corpus summary. Length–metric correlations are weak ($|r| < 0.17$), confirming that length differences do not drive findings.

	Classical (DLC)	Popular (LMD)
Total pieces	1,280	1,569
Composers/Genres	36 composers	12 genres
Time span	1680–1920	1950–2020
Piece length (beats)	3–1,594	8–456
Median length (beats)	156	128
Key annotation	Expert	Algorithmic (K-S)
Pitch spelling	Ground truth	Inferred (partitura)

A.2 Quality Filtering (LMD)

The LMD corpus was filtered in six stages to remove low-quality transcriptions and genre overlap (Table 2).

Table 2: Quality filtering stages. Genre filtering used Million Song Dataset metadata linked to LMD via track identifiers.

Filter	Threshold	Removed
Unique pitch classes	< 5	12,458
Pitch entropy	< 1.5 bits	8,234
Single pitch dominance	$> 50\%$ duration	5,891
Genre exclusion	Classical, Jazz, Blues, New Age	18,742
Pitch entropy (upper)	> 3.2 bits	272
Tonal focus	< 0.3	108
Remaining after filtering		1,569

A.3 Genre and Composer Variation

The heterogeneity visible in Figure 2 implies that aggregating all popular music into a single category masks meaningful organizational differences, and may partly explain the high within-tradition variance reported in the main text.

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Figure 1 reveals a progressive expansion from tight Baroque clustering toward greater Romantic and early 20th-century dispersion—suggesting that "stylistic period" captures a real shift in organizational strategy, not merely chronological grouping.

B. ROBUSTNESS AND IMPLEMENTATION

B.1 Non-Parametric Effect Sizes

Given the non-normal distributions in our data, we supplement Cohen's d with Cliff's delta (δ), a non-parametric effect size scaled to $[-1, 1]$. The results confirm the parametric findings and show that λ and focus provide the most reliable differentiation between traditions.

Table 3: Non-parametric effect sizes. Positive δ: Classical $>$ Popular; negative: Popular $>$ Classical.

Metric	Cliff's δ	Direction	Interpretation
λ (Path Length)	0.45	Cl $>$ Pop	Medium
Tonal Focus ($k = 3$)	-0.27	Pop $>$ Cl	Small-Medium
Fifth Dominance	0.15	Cl $>$ Pop	Small
Weight Entropy	0.05	Cl $>$ Pop	Negligible
Weight Kurtosis	0.34	Cl $>$ Pop	Small-Medium

B.2 Modulation Stratification

To test whether modulation rate confounds our findings, we stratified the classical corpus into terciles by modulation rate. Kurtosis and fifth dominance are stable across strata, confirming these metrics reflect organizational strategy rather than modulation frequency. The significant variation in λ is theoretically expected, since modulation is itself one mechanism for tonal traversal.

Table 4: Weight metrics stratified by modulation rate (classical corpus). Kurtosis and fifth dominance are stable; λ varies with modulation as expected.

Stratum	N	Kurtosis	5th Dom	λ
Low (< 0.02)	435	-0.08 ± 0.71	0.78 ± 0.10	2.16 ± 0.74
Medium	436	-0.06 ± 0.73	0.78 ± 0.09	2.35 ± 0.66
High (> 0.08)	434	-0.11 ± 0.70	0.79 ± 0.09	2.23 ± 0.67
Stability (p)		0.52	0.30	< 0.001

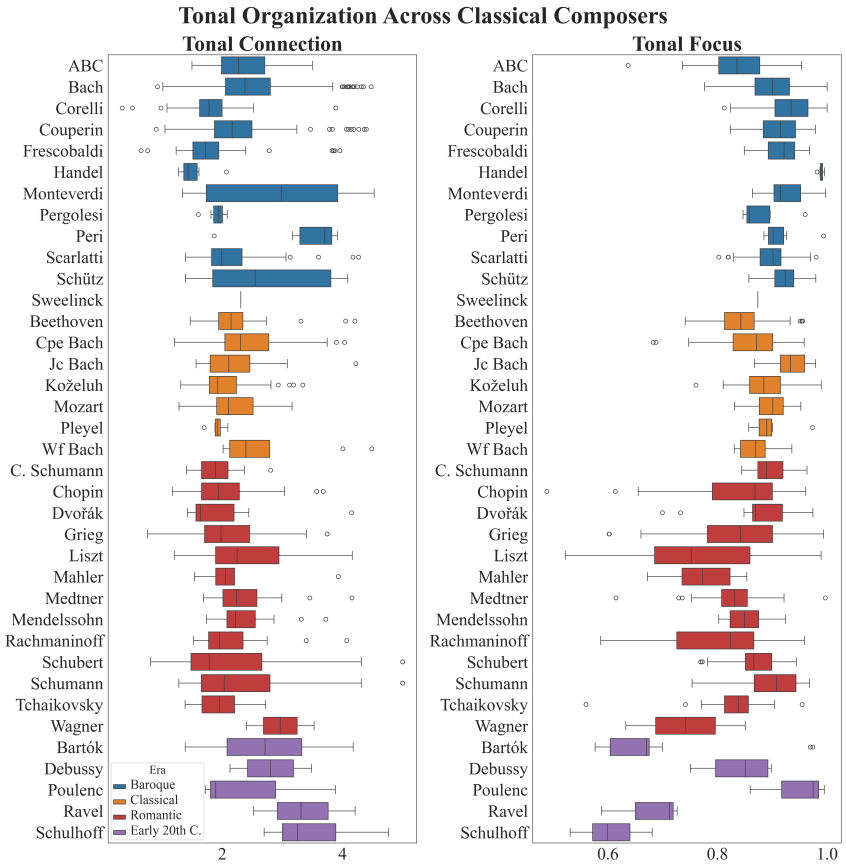

Figure 1: **Composer-level variation reveals historical trajectories within the classical tradition.** Box plots show tonal connection (left) and tonal focus (right) for 36 composers, colored by era. Baroque composers cluster at high focus with moderate-to-high connection; Romantic composers spread across a wider range; Early 20th-century composers show the greatest diversity, with some (Poulenc) maintaining traditional profiles while others (Schulhoff, Bartók) push toward low focus and high connection.

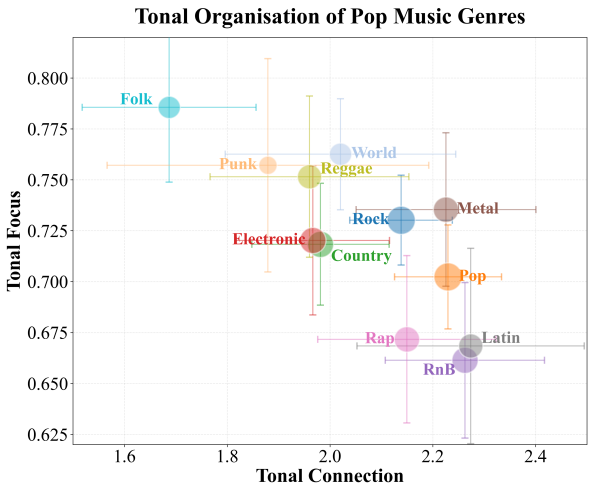

Figure 2: **Genre-level variation within popular music.** Each point represents a genre's mean position in the two-dimensional space; error bars indicate standard deviation. Folk occupies the high-focus, low-connection region, while Latin and World show high connection with lower focus. The spread confirms that "popular music" encompasses diverse organizational strategies.

B.3 Key Estimation Validation

Since MIDI key metadata is unreliable—over 75% of LMD files default to "C major"—we use Krumhansl-Schmuckler estimation for all popular music. The bottom rows of Table 5 show that both λ and focus are reasonably robust to ± 1 key estimation error, though focus exhibits greater sensitivity.

Table 5: Key distributions: raw MIDI metadata vs. K-S estimation (LMD). Bottom rows show robustness of λ and focus to ± 1 key shift.

Key	MIDI Metadata	K-S Estimated
C major	75.3%	14.7%
G major	3.2%	10.5%
D major	2.1%	8.7%
A major	1.8%	7.5%
F major	2.4%	6.6%
Other	15.2%	52.0%
Robustness to key shift (± 1 fifth):		
λ_K vs. $\lambda_{K\pm 1}$	$r = 0.76$	
Focus_K vs. $\text{Focus}_{K\pm 1}$	$r = 0.90$–0.94	

B.4 Representation and Temporal Robustness

To rule out artifacts of pitch spelling or percussion contamination, we replicated the analysis using 12-D pitch-class representations and compared full-piece vs. opening-window metrics. Both checks strengthen rather than weaken the reported differences.

Table 6: Robustness checks. *Top:* 12-D vs. 35-D comparison—larger 12-D effect sizes confirm that spelling noise and percussion attenuate rather than inflate differences. *Bottom:* Windowed analysis—classical λ is stable across windows while popular λ drops substantially in openings.

	35-D		12-D		
Metric	Classical	Pop	Classical	Pop	d (12-D)
λ	2.25	1.82	0.92	0.89	0.60
Fifth Dom	0.79	0.74	0.59	0.33	2.42
Windowed analysis (full-piece vs. opening):					
	Classical		Popular		
	Global	Window	Global	Window	
λ	2.25	2.14	1.82	1.30	
Change	-5%		-29%		

B.5 Correlation Structure

Table 7 reports pairwise correlations across all metrics, confirming that λ and focus capture partially independent dimensions.

Table 7: Correlation matrices: classical (above diagonal) and popular (below diagonal). λ and focus show partial independence ($r = -0.20$ classical, $r = -0.06$ popular); fifth dominance correlates with λ in both ($r \approx 0.51$).

	λ	Focus	5th Dom	Entropy	KL Div	Width
λ	—	-0.20	0.51	0.45	-0.45	0.28
Focus	-0.06	—	0.30	-0.42	0.42	-0.26
5th Dom	0.51	0.34	—	-0.26	0.26	-0.26
Entropy	0.43	-0.30	0.12	—	-1.00	0.84
KL Div	-0.43	0.30	-0.12	-1.00	—	-0.84
Width	0.16	-0.20	-0.10	0.71	-0.71	—

B.6 Pitch Spelling Algorithm

For MIDI files lacking ground-truth pitch spellings, we use the `partitura` library's HMM-based spelling algorithm [1], which estimates the global key, defines transition probabilities based on voice-leading preferences, and applies the Viterbi algorithm to infer spellings. Validation on the DLC achieves 94.2% agreement with expert spellings at the note level.

B.7 TDM Fitting Procedure

The Tonal Diffusion Model parameters (λ, $\mathbf{w}$) are estimated per piece through maximum likelihood optimization. For each piece with observed pitch-class distribution $\mathbf{d}$ and tonal center c:

1. Initialize $\lambda = 2.0$ and $\mathbf{w}$ as uniform over six intervals.
2. Compute the predicted distribution $\mathbf{p}(\lambda, \mathbf{w}, c)$ by convolving the random walk kernel with the tonic position.
3. Optimize $\{\lambda, \mathbf{w}\}$ to maximize $\sum_i d_i \log p_i$ using L-BFGS-B with bounds $\lambda \in [0.1, 10]$ and $w_i \in [0.01, 0.99]$.
4. Normalize $\mathbf{w}$ to sum to 1 after optimization.

Convergence criterion: gradient norm $< 10^{-6}$ or 1000 iterations. Over 99% of pieces converge within 100 iterations.

C. REFERENCES

[1] C. Cancino-Chacón, M. Grachten, W. Goebl, and G. Widmer, "Partitura: A python package for symbolic music processing," in *Proceedings of the Music Encoding Conference (MEC)*, College Park, MD, USA, 2018.